Semester: Fall 26
Instructor: Zitan Chen

MCE6001: Random Processes
Assignment 1
Date due: September 30, 5:00 p.m.

Brief sample solutions are given below. *Complete* solutions are required for *full* marks.

Problem 1. (10 pts) Let $X_1, X_2, \dots$ be independent random variables with $\mathrm{E}[X_i] = 0, \mathrm{E}[X_i^2] < \infty$, and define

$$S_n = \sum_{i=1}^{n} X_i.$$

Prove that $(S_n)_{n\geq 1}$ converges in mean square if and only if

$$\sum_{i=1}^{\infty} \mathrm{Var}(X_i) < \infty.$$

Hint: Apply the Cauchy criterion of mean square convergence.

For $m > n$, independence and the zero-mean assumption imply

$$\mathrm{E}[|S_m - S_n|^2] = \mathrm{E}\left[\left(\sum_{i=n+1}^{m} X_i\right)^2\right] = \sum_{i=n+1}^{m} \mathrm{Var}(X_i).$$

Thus, by the Cauchy criterion, $(S_n)_{n\geq 1}$ converges in mean square if and only if $\lim_{m,n\to\infty} \sum_{i=n+1}^{m} \mathrm{Var}(X_i) = 0$, i.e., the tails of the nonnegative series $\sum_{i=1}^{\infty} \mathrm{Var}(X_i)$ tend to zero, which is equivalent to that series being finite.

Problem 2. (30 pts) Let $X_1, X_2, \dots$ be independent random variables with

$$P\{X_i = 1\} = P\{X_i = -1\} = \frac{1}{2} \text{ for } i \geq 1.$$

Define

$$S_n = \sum_{i=1}^{n} X_i, \qquad V_n = \frac{S_n}{\sqrt{n}},$$

and write $\Phi_Z(u) = \mathrm{E}[e^{juZ}]$, $u \in \mathbb{R}$, for the characteristic function of a random variable Z.

(a) Find Φ_{X_i}, Φ_{S_n}, and Φ_{V_n}. *Hint: Euler's formula gives* $\cos u = \frac{e^{ju}+e^{-ju}}{2}$, $u \in \mathbb{R}$.

We have

$$\Phi_{X_i}(u) = \frac{e^{ju} + e^{-ju}}{2} = \cos u.$$

By independence and scaling of random variables,

$$\Phi_{S_n}(u) = (\cos u)^n, \qquad \Phi_{V_n}(u) = \left(\cos\frac{u}{\sqrt{n}}\right)^n.$$

(b) For each fixed $u \in \mathbb{R}$, determine whether $\Phi_{S_n}(u)$ and $\Phi_{V_n}(u)$ converge as $n \to \infty$, and find their limits when they exist.

作业 1 官方解答 P1

For Φ_{S_n}, the behavior depends on u:

- If $u = 2k\pi$ with $k \in \mathbb{Z}$, then $\Phi_{S_n}(u) = 1$.
- If $u \neq k\pi$ with $k \in \mathbb{Z}$, then $|\cos u| < 1$, so $\Phi_{S_n}(u) \to 0$.
- If $u = (2k+1)\pi$ with $k \in \mathbb{Z}$, then $\Phi_{S_n}(u) = (-1)^n$, so the limit does not exist.

In particular, $(\Phi_{S_n})_{n\geq 1}$ has no pointwise limit on all of $\mathbb{R}$.

Note that $\mathrm{E}[X_i] = 0$ and $\mathrm{Var}(X_i) = 1$. For each fixed $t \in \mathbb{R}$, CLT gives $\Phi_{V_n}(t) \longrightarrow e^{-t^2/2}$.

(c) Determine whether $(S_n)_{n\geq 1}$ converge almost surely, in mean square, in probability, and in distribution. Identify the limit whenever convergence holds, and justify all convergence and nonconvergence claims.

By Part (b), the sequence $(S_n)_{n\geq 1}$ does not converge in distribution: distributional convergence would imply convergence of its characteristic functions at every t, whereas $\Phi_{S_n}(\pi) = (-1)^n$ does not converge. Because almost-sure, mean-square, and probability convergence each imply distributional convergence, $(S_n)_{n\geq 1}$ converges in none of the four senses.

Problem 3. (20 pts)

(a) Let Z be a random variable with finite mean, whose moment-generating function $M_Z(t) = \mathrm{E}[e^{tZ}]$ is finite for every $t \geq 0$. Using Markov's inequality, prove the following version of Chernoff bound:

$$P\{Z \geq \mathrm{E}[Z] + c\} \leq e^{-t(\mathrm{E}[Z]+c)} M_Z(t), \qquad c \geq 0, \quad t \geq 0.$$

For $t > 0$, the function $z \mapsto e^{tz}$ is strictly increasing. Applying Markov's inequality to the nonnegative random variable e^{tZ} gives

$$\begin{aligned} P\{Z \geq \mathrm{E}[Z] + c\} &= P\{e^{tZ} \geq e^{t(\mathrm{E}[Z]+c)}\} \\ &\leq \frac{\mathrm{E}[e^{tZ}]}{e^{t(\mathrm{E}[Z]+c)}} \\ &= e^{-t(\mathrm{E}[Z]+c)} M_Z(t). \end{aligned}$$

(b) Let $Z \sim \mathrm{N}(\mu, \sigma^2)$ where $\sigma > 0$, i.e., Z is a Gaussian random variable with mean μ and variance σ^2. For each $c \geq 0$, find the optimized Chernoff bound in Part (a) and give an optimizing t.

Since the Chernoff bound in Part (a) holds for every $t \geq 0$, taking the infimum yields

$$P\{Z \geq \mathrm{E}[Z] + c\} \leq \inf_{t\geq 0} e^{-t(\mathrm{E}[Z]+c)} M_Z(t).$$

Since $\mathrm{E}[Z] = \mu$ and

$$M_Z(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right),$$

we have

$$B_Z(c) := \inf_{t\geq 0} \exp\left(\frac{\sigma^2 t^2}{2} - ct\right).$$

The exponent is minimized at $t_* = c/\sigma^2 \geq 0$. Thus, $P\{Z \geq \mu + c\} \leq B_Z(c) = \exp\left(-c^2/(2\sigma^2)\right)$.

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Problem 4. (20 pts) Let $X_n \xrightarrow{a.s.} X$ and $g : \mathbb{R} \to \mathbb{R}$ be a continuous function, and define

$$h(x) = \begin{cases} -1, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

For each sequence below, must an almost-sure limit exist? If yes, identify the limit and justify your answer. If no, give a counterexample in which the transformed sequence does not converge almost surely.

(a) $(g(X_n))_{n\geq 1}$. *Hint: Recall the definition of continuous functions.*

Yes. Let $A = \{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}$, so that $P(A) = 1$. Fix $\omega \in A$ and $\varepsilon > 0$. By continuity of g at $X(\omega)$, there exists $\delta > 0$ such that

$$|x - X(\omega)| < \delta \implies |g(x) - g(X(\omega))| < \varepsilon.$$

Since $X_n(\omega) \to X(\omega)$, there exists N such that

$$|X_n(\omega) - X(\omega)| < \delta \qquad \text{for all } n \geq N.$$

Consequently,

$$|g(X_n(\omega)) - g(X(\omega))| < \varepsilon \qquad \text{for all } n \geq N.$$

Thus, $g(X_n(\omega)) \to g(X(\omega))$ on the probability-one event A. It follows that

$$g(X_n) \xrightarrow{a.s.} g(X).$$

(b) $(h(X_n))_{n\geq 1}$.

No. Take the constant random variables

$$X_n = \frac{(-1)^n}{n}, \qquad X = 0.$$

Then $X_n \to X$ almost surely, but

$$h(X_n) = (-1)^n,$$

which does not converge.

Problem 5. (20 pts) Let Θ be uniformly distributed on the interval $[0, 2\pi)$. For each sequence below, determine whether it converges almost surely, in mean square, in probability, and in distribution. Identify the limit whenever convergence holds, and justify all convergence and nonconvergence claims.

(a) $X_n = \cos(n\Theta)$, $n \geq 1$.

Convergence in distribution: Let $U_n = n\Theta \pmod{2\pi} \in [0, 2\pi)$. Since $n\Theta$ is uniform on $[0, 2\pi n)$, for $0 \leq a < b < 2\pi$,

$$P\{a \leq U_n < b\} = \sum_{k=0}^{n-1} P\{2\pi k + a \leq n\Theta < 2\pi k + b\} = n\frac{b-a}{2\pi n} = \frac{b-a}{2\pi}.$$

Therefore, U_n is uniformly distributed on $[0, 2\pi)$. Thus, by periodicity, all X_n have the same distribution as $\cos\Theta$, and

$$X_n \xrightarrow{d} \cos\Theta.$$

Convergence in mean square: For every nonzero $k \geq 1$, we have

$$\int_0^{2\pi} \cos(k\theta) d\theta = \frac{\sin(k\theta)}{k}\Big|_0^{2\pi} = 0.$$

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Since $\Theta \sim \mathrm{U}[0, 2\pi]$, the trigonometric identities $\cos^2 u = (1 + \cos(2u))/2$ and $\cos u \cos v = (\cos(u-v) + \cos(u+v))/2$ give

$$\begin{aligned} \mathrm{E}[X_n^2] &= \frac{1}{4\pi}\int_0^{2\pi} \big(1 + \cos(2n\theta)\big)\, d\theta = \frac{1}{2}, \\ \mathrm{E}[X_n X_m] &= \frac{1}{4\pi}\int_0^{2\pi} \big(\cos((n-m)\theta) + \cos((n+m)\theta)\big)\, d\theta = 0 \qquad (m \neq n). \end{aligned}$$

The second equality uses that $n - m$ and $n + m$ are nonzero integers when $m, n \geq 1$ and $m \neq n$. Consequently, for all distinct m, n,

$$\mathrm{E}[(X_n - X_m)^2] = \mathrm{E}[X_n^2] + \mathrm{E}[X_m^2] - 2\,\mathrm{E}[X_n X_m] = 1.$$

Therefore, $(X_n)_{n\geq 1}$ does not converge in mean square, by the Cauchy criterion.

Convergence in probability and a.s.: For $m \neq n$, let $A = \{|X_n - X_m| > 1/2\}$. On A^c, the squared difference is at most $1/4$; on A, it is at most 4, since $|X_n - X_m| \leq 2$. Consequently,

$$\begin{aligned} 1 = \mathrm{E}[(X_n - X_m)^2] &= \mathrm{E}[(X_n - X_m)^2 \mathbf{1}_{A^c}] + \mathrm{E}[(X_n - X_m)^2 \mathbf{1}_A] \\ &\leq \frac{1}{4} P(A^c) + 4P(A) \\ &\leq \frac{1}{4} + 4P(A). \end{aligned}$$

Hence,

$$P\left\{|X_n - X_m| > \frac{1}{2}\right\} \geq \frac{3}{16} \qquad (m \neq n).$$

Thus, $(X_n)_{n\geq 1}$ does not in probability by the Cauchy criterion, so it cannot converge in probability or almost surely.

In summary, $(X_n)_{n\geq 1}$ converges only in distribution among the four specified senses.

(b) $Y_n = |1 - \Theta/\pi|^n$, $n \geq 1$.

Set $U = |1 - \Theta/\pi|$, so that $0 \leq U \leq 1$. For $0 \leq u \leq 1$, uniformity of Θ gives

$$\begin{aligned} P\{U \leq u\} &= P\{\pi(1-u) \leq \Theta \leq \pi(1+u)\} \\ &= \frac{\pi(1+u) - \pi(1-u)}{2\pi} = u. \end{aligned}$$

Hence, $U \sim \mathrm{U}[0, 1]$.

Moreover,

$$P\{U = 1\} = P\{\Theta \in \{0, 2\pi\}\} = 0.$$

On the probability-one event $\{0 \leq U < 1\}$, each $U(\omega)$ is a fixed number strictly less than 1, so

$$Y_n(\omega) = U(\omega)^n \longrightarrow 0.$$

Therefore, $Y_n \xrightarrow{a.s.} 0$.

Also,

$$\mathrm{E}[|Y_n - 0|^2] = \int_0^1 u^{2n}\, du = \frac{1}{2n+1} \longrightarrow 0.$$

Hence, $Y_n \to 0$ in mean square as well. Almost-sure convergence implies convergence in probability, which implies convergence in distribution.

Thus, $(Y_n)_{n\geq 1}$ converges to 0 in all four senses.

作业 1 官方解答 P4

Semester: Fall 26
Instructor: Zitan Chen

MCE6001: Random Processes
Assignment 2
Date due: October 28, 5:00 p.m.

Brief sample solutions are given below. *Complete* solutions are required for *full* marks.

Problem 1. (30 pts) Let X and Y be jointly Gaussian random variables with mean zero and covariance matrix

$$\mathrm{Cov}\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 6 & 18 \end{pmatrix}.$$

You may express your answers using the standard normal CDF:

$$\Phi(u) = \int_{-\infty}^{u} \frac{1}{\sqrt{2\pi}} e^{-s^2/2}\, ds.$$

(a) Find $P\{|X - 1| \geq 2\}$.

(b) Find the conditional density of X given that $Y = 3$. You can either write out the density in full, or describe it as a well-known density with specified parameter values.

(c) Find $P\{|X - \mathrm{E}[X \mid Y]| \geq 1\}$.

The covariance matrix gives

$$\mathrm{Var}(X) = 4, \qquad \mathrm{Var}(Y) = 18, \qquad \mathrm{Cov}(X, Y) = 6.$$

(a) Since $X \sim \mathrm{N}(0, 4)$, the variable $X/2$ is standard normal. Also,

$$\{|X - 1| \geq 2\} = \{X \leq -1\} \cup \{X \geq 3\}.$$

These events are disjoint, so

$$P(|X - 1| \geq 2) = \Phi\left(-\frac{1}{2}\right) + 1 - \Phi\left(\frac{3}{2}\right).$$

(b) For jointly Gaussian variables, the conditional distribution is Gaussian, with mean

$$\mathrm{E}[X] + \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(Y)}(y - \mathrm{E}[Y]) = \frac{y}{3}$$

and variance

$$\mathrm{Var}(X) - \frac{\mathrm{Cov}(X, Y)^2}{\mathrm{Var}(Y)} = 4 - \frac{36}{18} = 2.$$

Therefore,

$$(X \mid Y = 3) \sim \mathrm{N}(1, 2).$$

Equivalently, the density of $(X \mid Y = 3)$ is

$$f_{X|Y}(x \mid 3) = \frac{1}{\sqrt{4\pi}} \exp\left(-\frac{(x-1)^2}{4}\right), \qquad x \in \mathbb{R}.$$

作业 2 官方解答 P1

(c) From part (b), $\mathrm{E}[X \mid Y] = Y/3$. Define

$$Z = X - \frac{Y}{3}.$$

As a linear combination of jointly Gaussian variables, Z is also Gaussian. Its mean is zero, and

$$\mathrm{Var}(Z) = \mathrm{Var}\left(X - \frac{Y}{3}\right) = 4 + \frac{18}{9} - \frac{2}{3} \cdot 6 = 2.$$

It follows that $Z/\sqrt{2} \sim \mathrm{N}(0, 1)$, and thus we obtain

$$P\left(|X - \mathrm{E}[X \mid Y]| \geq 1\right) = P(|Z| \geq 1) = 2\left[1 - \Phi\left(\frac{1}{\sqrt{2}}\right)\right].$$

Problem 2. (30 pts) Let $(Y_1, Y_2, Y_3, X)^T$ be a mean-zero random vector with covariance matrix

$$\begin{pmatrix} 1 & 1/2 & 1/2 & 0 \\ 1/2 & 1 & 1/2 & 1/4 \\ 1/2 & 1/2 & 1 & 1/4 \\ 0 & 1/4 & 1/4 & 1 \end{pmatrix}.$$

Let $\tilde{Y}_1, \tilde{Y}_2, \tilde{Y}_3$ be the linear innovations sequence for Y_1, Y_2, Y_3. Write

$$\mathbf{Y} = (Y_1, Y_2, Y_3)^T, \qquad \tilde{\mathbf{Y}} = (\tilde{Y}_1, \tilde{Y}_2, \tilde{Y}_3)^T.$$

(a) Find the matrix A such that $\tilde{\mathbf{Y}} = A\mathbf{Y}$.

The linear innovations sequence is constructed recursively by

$$\tilde{Y}_1 = Y_1 - \mathrm{E}[Y_1], \text{ and } \tilde{Y}_j = Y_j - \sum_{i=1}^{j-1} \frac{\mathrm{Cov}(Y_j, \tilde{Y}_i)}{\mathrm{Var}(\tilde{Y}_i)} \tilde{Y}_i \text{ for } j \geq 2.$$

First,

$$\tilde{Y}_1 = Y_1 - \mathrm{E}[Y_1] = Y_1, \qquad \mathrm{Var}(\tilde{Y}_1) = 1.$$

Next,

$$\tilde{Y}_2 = Y_2 - \frac{\mathrm{Cov}(Y_2, \tilde{Y}_1)}{\mathrm{Var}(\tilde{Y}_1)} \tilde{Y}_1 = Y_2 - \frac{1}{2} Y_1, \qquad \mathrm{Var}(\tilde{Y}_2) = 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}.$$

For the third innovation,

$$\mathrm{Cov}(Y_3, \tilde{Y}_1) = \frac{1}{2},$$

and

$$\mathrm{Cov}(Y_3, \tilde{Y}_2) = \mathrm{Cov}\left(Y_3, Y_2 - \frac{1}{2} Y_1\right) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

Hence,

$$\begin{aligned} \tilde{Y}_3 &= Y_3 - \frac{1}{2}\tilde{Y}_1 - \frac{1/4}{3/4}\tilde{Y}_2 \\ &= Y_3 - \frac{1}{2} Y_1 - \frac{1}{3}\left(Y_2 - \frac{1}{2} Y_1\right) \\ &= Y_3 - \frac{1}{3} Y_1 - \frac{1}{3} Y_2. \end{aligned}$$

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Therefore,

$$A = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ -\frac{1}{3} & -\frac{1}{3} & 1 \end{pmatrix}.$$

(b) Find $\mathrm{Cov}(\tilde{\mathbf{Y}})$ and the cross-covariance

$$\mathrm{Cov}(X, \tilde{\mathbf{Y}}) = \mathrm{E}[X\tilde{\mathbf{Y}}^T].$$

The linear innovations sequence are pairwise uncorrelated by construction. Their first two variances, $\mathrm{Var}(\tilde{Y}_1)$ and $\mathrm{Var}(\tilde{Y}_2)$, were computed in Part (a). For the third, since $\tilde{Y}_3 = Y_3 - \tilde{Y}_1/2 - \tilde{Y}_2/3$, orthogonality gives $\mathrm{Var}(\tilde{Y}_3) = \mathrm{Var}(\tilde{Y}_1)/4 + \mathrm{Var}(\tilde{Y}_2)/9 + \mathrm{Var}(\tilde{Y}_3)$. It follows that

$$\mathrm{Var}(\tilde{Y}_3) = 1 - \frac{1}{4} - \frac{3/4}{9} = \frac{2}{3}.$$

Thus,

$$\mathrm{Cov}(\tilde{\mathbf{Y}}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{4} & 0 \\ 0 & 0 & \frac{2}{3} \end{pmatrix}.$$

The cross-covariances are

$$\begin{aligned} &\mathrm{Cov}(X, \tilde{Y}_1) = \mathrm{Cov}(X, Y_1) = 0, \\ &\mathrm{Cov}(X, \tilde{Y}_2) = \mathrm{Cov}\left(X, Y_2 - \frac{Y_1}{2}\right) = \frac{1}{4} - \frac{1}{2} \cdot 0 = \frac{1}{4}, \\ &\mathrm{Cov}(X, \tilde{Y}_3) = \mathrm{Cov}\left(X, Y_2 - \frac{Y_1}{3} - \frac{\tilde{Y}_2}{3}\right) = \frac{1}{4} - \frac{1}{3} \cdot 0 - \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{6}. \end{aligned}$$

Consequently,

$$\mathrm{Cov}(X, \tilde{\mathbf{Y}}) = \left(0, \frac{1}{4}, \frac{1}{6}\right).$$

(c) Find $a, b, c \in \mathbb{R}$ that minimize $\mathrm{E}[(X - a\tilde{Y}_1 - b\tilde{Y}_2 - c\tilde{Y}_3)^2]$.

By the orthogonality principle, the optimal error

$$e = X - a\tilde{Y}_1 - b\tilde{Y}_2 - c\tilde{Y}_3$$

must satisfy

$$\mathrm{E}[e\tilde{Y}_i] = 0, \qquad i = 1, 2, 3.$$

Because the innovations are pairwise uncorrelated, these equations yield:

$$a = \frac{\mathrm{Cov}(X, \tilde{Y}_1)}{\mathrm{Var}(\tilde{Y}_1)} = 0, \qquad b = \frac{\mathrm{Cov}(X, \tilde{Y}_2)}{\mathrm{Var}(\tilde{Y}_2)} = \frac{1/4}{3/4} = \frac{1}{3}, \qquad c = \frac{\mathrm{Cov}(X, \tilde{Y}_3)}{\mathrm{Var}(\tilde{Y}_3)} = \frac{1/6}{2/3} = \frac{1}{4}.$$

Problem 3. (10 pts) Let $X = \frac{1}{1+U}$, where $U \sim \mathrm{U}[0, 1]$ is uniformly distributed over the interval $[0, 1]$. Find the conditional expectation $\mathrm{E}[X|U]$ and its MSE $\mathrm{E}[(X - \mathrm{E}[X|U])^2]$.

Observing $U = u$ determines X exactly, so

$$\mathrm{E}[X|U = u] = \mathrm{E}\left[\frac{1}{1+U}\middle| U = u\right] = \frac{1}{1+u}, \quad u \in [0, 1],$$

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and therefore,

$$\mathrm{E}[X|U] = \frac{1}{1+U}.$$

Consequently,

$$\mathrm{E}\left[(X - \mathrm{E}[X \mid U])^2\right] = 0.$$

Problem 4. (10 pts) Let $Y = X + N$, where X and N are independent, $N \sim \mathrm{N}(0, \sigma^2)$, and $X \sim \mathrm{Exp}(\lambda)$. Assume $\lambda > 0$ and $\sigma^2 > 0$. Find $\hat{\mathrm{E}}[X|Y]$ and the mean square error for estimating X by $\hat{\mathrm{E}}[X|Y]$.

Hint: Recall that $\mathrm{E}[X] = 1/\lambda$ *and* $\mathrm{Var}(X) = 1/\lambda^2$.

Note that

$$\mathrm{E}[X] = \mathrm{E}[Y] = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

By independence,

$$\mathrm{Var}(Y) = \frac{1}{\lambda^2} + \sigma^2, \qquad \mathrm{Cov}(X, Y) = \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

Write $\hat{\mathrm{E}}[X|Y] = a + bY$. Then the orthogonality conditions are

$$\mathrm{E}[X - a - bY] = 0, \qquad \mathrm{E}[(X - a - bY)Y] = 0.$$

The first condition yields $a = \mathrm{E}[X] - b\,\mathrm{E}[Y]$. Substituting this into the second gives

$$\mathrm{E}[(X - \mathrm{E}[X])Y] = b\mathrm{E}[(Y - \mathrm{E}[Y])Y].$$

By properties of covariances, we further obtain

$$b = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(Y)} = \frac{1}{1 + \lambda^2\sigma^2}.$$

Therefore,

$$a = \mathrm{E}[X] - b\mathrm{E}[Y] = \frac{1}{\lambda} - \frac{1}{\lambda(1 + \lambda^2\sigma^2)} = \frac{\lambda\sigma^2}{1 + \lambda^2\sigma^2},$$

and

$$\hat{\mathrm{E}}[X|Y] = a + bY = \frac{Y + \lambda\sigma^2}{1 + \lambda^2\sigma^2}.$$

The minimum MSE is

$$\begin{aligned} \mathrm{E}[(X - \hat{\mathrm{E}}[X|Y])^2] &= \mathrm{Var}(X) - \frac{\mathrm{Cov}(X, Y)^2}{\mathrm{Var}(Y)} \\ &= \frac{1}{\lambda^2} - \frac{1/\lambda^4}{1/\lambda^2 + \sigma^2} \\ &= \frac{\sigma^2}{1 + \lambda^2\sigma^2}. \end{aligned}$$

Problem 5. (20 pts) For $k \geq 0$, consider the state and observation (scalar) sequences defined by:

$$\begin{aligned} x_{k+1} &= fx_k + w_k, \\ y_k &= x_k + v_k, \end{aligned}$$

where $f \in \mathbb{R}$ and $x_0 \sim \mathrm{N}(0, \sigma^2)$ with $\sigma^2 > 0$. Assume that $x_0, w_0, v_0, w_1, v_1, \dots$ are independent and that $w_k, v_k \sim \mathrm{N}(0, 1)$ for every $k \geq 0$.

作业 2 官方解答 P4

Write down the Kalman filter equations for recursively computing the estimates $\hat{x}_{k+1|k}$, the (scalar) gains K_k, and the sequence of the variances of the errors (for brevity, denote $\sigma_{k+1}^2 := \mathrm{Cov}(x_{k+1} - \hat{x}_{k+1|k})$ for the covariance of error instead of $\Sigma_{k+1|k}$).

Before observing y_0, since $x_0 \sim \mathrm{N}(0, \sigma^2)$, the initialization is

$$\hat{x}_{0|-1} = 0, \qquad \sigma_0^2 = \sigma^2.$$

Information Update: For $k \geq 0$, define the measurement innovation

$$\nu_k = y_k - \hat{x}_{k|k-1}.$$

Let $e_k = x_k - \hat{x}_{k|k-1}$. Then from the measurement $y_k = x_k + v_k$, we have $\nu_k = e_k + v_k$. Since v_k is independent of the prior estimation error e_k,

$$\mathrm{Var}(\nu_k) = \sigma_k^2 + 1, \qquad \mathrm{Cov}(e_k, \nu_k) = \sigma_k^2.$$

The Kalman gain is therefore

$$K_k = \frac{f\sigma_k^2}{\sigma_k^2 + 1}.$$

The updated estimate is

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + \frac{\sigma_k^2}{\sigma_k^2 + 1}(y_k - \hat{x}_{k|k-1}).$$

The updated error satisfies

$$x_k - \hat{x}_{k|k} = (1 - K_k)e_k - K_k v_k.$$

Hence, the updated error variance is

$$\mathrm{Var}(x_k - \hat{x}_{k|k}) = (1 - K_k)^2\sigma_k^2 + K_k^2 = \frac{\sigma_k^2}{1 + \sigma_k^2}.$$

Time Update: The next-state prediction uses $x_{k+1} = fx_k + w_k$, where w_k is independent of the updated estimation error. Thus,

$$\hat{x}_{k+1|k} = \hat{\mathrm{E}}[x_{k+1} \mid y_0, \dots, y_k] = f\hat{x}_{k|k}, \qquad \sigma_{k+1}^2 = 1 + \frac{f^2\sigma_k^2}{1 + \sigma_k^2}.$$

Equivalently, the estimate can be updated directly as

$$\hat{x}_{k+1|k} = f\hat{x}_{k|k-1} + K_k(y_k - \hat{x}_{k|k-1}).$$

作业 2 官方解答 P5

考点 1 | 均方收敛与 Cauchy 准则（作业1-P1）

- **记背** m.s. 定义 $E[(X_n-X)^2]\to 0$；Cauchy 准则 $\lim_{m,n\to\infty}E[(X_n-X_m)^2]=0$ ⇔ m.s. 收敛（L2 完备，不需知极限）；独立零均值 $\Rightarrow E[(S_n-S_m)^2]=\sum_{i=m+1}^{n}\mathrm{Var}(X_i)$；结论 S_n m.s. 收敛 ⇔ $\sum_{i=1}^{\infty}\mathrm{Var}(X_i)<\infty$。
- **套路** 展开平方→独立/零均值杀交叉项→双向套 Cauchy。
- **易错** 只证单向；忘 $E[X_i]=0\Rightarrow E[X_i^2]=\mathrm{Var}(X_i)$。

考点 2 | 特征函数与四种收敛判定（作业1-P2、P5）

- **记背** $\Phi_Z(u)=E[e^{juZ}]$；独立和⇒相乘；$\Phi_{aX+b}(u)=e^{jub}\Phi_X(au)$；$\cos u=\frac{e^{ju}+e^{-ju}}{2}$⇒Rademacher $\Phi_{S_n}=(\cos u)^n$；$(\cos\frac{u}{\sqrt{n}})^n\to e^{-u^2/2}$（CLT 原型）；Lévy 连续性定理（d⇔Φ 逐点→0 处连续）；收敛链 $a.s.\Rightarrow p\Rightarrow d$、$m.s.\Rightarrow p\Rightarrow d$，反之均不成立；有界+p⇒m.s.；p⇒存在 a.s. 子列；d 且极限常数⇒p。
- **判定清单**：a.s. 看增量/逐点（稠密振荡）；m.s. 看 Cauchy/二阶矩；p 看定义界或借 d；d 看特征函数/CDF/**同分布⇒d 收敛到该分布。**
- **易错** $(\cos u)^n$ 在 $u=(2k+1)\pi$ 为 $(-1)^n$ 不收敛⇒否 d 从而否 p；$\cos(n\Theta)$ 依分布**是**收敛的。

考点 3 | 尾概率界（作业1-P3）

- **记背** Markov $P\{Y\ge a\}\le E[Y]/a$（$Y\ge 0$）；Chebyshev $P\{|X-\mu|\ge\epsilon\}\le\sigma^2/\epsilon^2$；Chernoff $P\{Z\ge E[Z]+c\}\le e^{-t(E[Z]+c)}M_Z(t)$；高斯 $M_Z(t)=e^{\mu t+\sigma^2t^2/2}$⇒最优界 $e^{-c^2/(2\sigma^2)}$、$t^*=c/\sigma^2$。
- **套路** 事件指数化（单调）→Markov→对 t 求导/配方优化。
- **易错** Markov 要求非负；验证 $t^*\ge 0$；下尾需 $t\le 0$。

考点 4 | 函数变换的收敛（作业1-P4）

- **记背** 连续映射定理：g 连续⇒a.s./p 收敛保持；d 版本需 $P\{X\in D_g\}=0$。
- **反例模板**：$X_n=(-1)^n/n\xrightarrow{a.s.}0$，阶跃 $h\Rightarrow h(X_n)=(-1)^n$ 不收敛。
- **易错** 反例的极限点必须落在 g 的间断点上。

考点 5 | 联合高斯（作业2-P1）

- **记背** $E[X|Y=y]=\mu_X+\frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}(y-\mu_Y)$；$\mathrm{Var}(X|Y)=\mathrm{Var}(X)-\frac{\mathrm{Cov}(X,Y)^2}{\mathrm{Var}(Y)}$；不相关⇔独立；误差 $X-E[X|Y]\perp$ 观测⇒独立；条件方差与 y 无关；$\Phi(.5)=.6915$，$\Phi(1)=.8413$，$\Phi(1.5)=.9332$，$\Phi(2)=.9772$，$\Phi(1/\sqrt{2})=.7602$。
- **套路** 协方差矩阵读均值方差→条件公式→误差概率化单变量高斯标准化。
- **易错** $\mathrm{Var}=4\Rightarrow\sigma=2$，别把方差当标准差。

考点 6 | 新息序列与正交 MMSE（作业2-P2）

- **记背** $\tilde{Y}_k=Y_k-\widehat{E}[Y_k|Y_1..Y_{k-1}]$；$\mathrm{Cov}(\tilde{Y})$ 对角，$\mathrm{Var}(\tilde{Y}_k)=\mathrm{Var}(Y_k)-$预测方差；$\widehat{E}[X|Y_1..Y_n]=\sum_i\frac{\mathrm{Cov}(X,\tilde{Y}_i)}{\mathrm{Var}(\tilde{Y}_i)}\tilde{Y}_i$；$\mathrm{MSE}=\mathrm{Var}(X)-\sum_i\frac{\mathrm{Cov}(X,\tilde{Y}_i)^2}{\mathrm{Var}(\tilde{Y}_i)}$；$\begin{pmatrix}1&r\\r&1\end{pmatrix}^{-1}=\frac{1}{1-r^2}\begin{pmatrix}1&-r\\-r&1\end{pmatrix}$。
- **套路** Gram–Schmidt 逐步减预测→对角化→系数逐项 Cov/Var。
- **易错** 2×2 逆矩阵符号；交叉协方差逐项展开别漏项。

考点 7 | 条件期望（作业2-P3 + 性质全单）

- **记背** $X=g(U)\Rightarrow E[X|U]=X$、MSE=0；$E[E[X|Y]]=E[X]$；取出已知 $E[g(Y)X|Y]=g(Y)E[X|Y]$；塔式 $E[E[X|Y,Z]|Y]=E[X|Y]$；全方差 $\mathrm{Var}(X)=E[\mathrm{Var}(X|Y)]+\mathrm{Var}(E[X|Y])$；$\mathrm{Var}(X|Y)=E[X^2|Y]-(E[X|Y])^2$；正交 $E[(X-E[X|Y])h(Y)]=0$。
- **套路** 先看可测性（是否观测的函数）→否则用联合密度积分或性质链。

考点 8 | 线性 MMSE 与精确 MMSE（作业2-P4）

- **记背** $\widehat{E}[X|Y]=\mu_X+\frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}(Y-\mu_Y)$；$\mathrm{MSE}=\mathrm{Var}(X)(1-\rho^2)$；$\mathrm{Cov}(X,X+N)=\mathrm{Var}(X)$；截断正态均值 $m+\sigma\frac{\phi(m/\sigma)}{\Phi(m/\sigma)}$（Exp 先验×高斯似然，$m=y-\lambda\sigma^2$）。
- **套路** 带帽 = 只要一二阶矩；不带帽 = Bayes 后验均值（配方认分布）。
- **易错** 帽与不帽区分；独立和方差相加、协方差只留自身项。

考点 9 | Kalman 滤波（作业2-P5）

- **记背** 标量模板 $K=\frac{\Sigma^-}{\Sigma^-+R}$；$\hat{x}^+=\hat{x}^-+K(y-\hat{x}^-)$；$\Sigma^+=(1-K)\Sigma^-$；$\hat{x}^-_{k+1}=F\hat{x}^+$；$\Sigma^-_{k+1}=F^2\Sigma^++Q$；作业特例 $K_k=\frac{\sigma_k^2}{1+\sigma_k^2}$、$\sigma^2_{k+1}=\frac{f^2\sigma_k^2}{1+\sigma_k^2}+1$；向量形式 $K=\Sigma^-H^T(H\Sigma^-H^T+R)^{-1}$、$\Sigma^+=(I-KH)\Sigma^-$、$\Sigma^-_{k+1}=F\Sigma^+F^T+Q$。
- **套路** 更新-预测两步写清 + 初值 $\hat{x}_{0|-1}=E[x_0]$、$\Sigma_0^-=\mathrm{Var}(x_0)$；可合并为一步递推。
- **易错** Q（过程噪声）与 R（观测噪声）别代反；下标自洽。

考点 10 | 平稳性与遍历性（课件 P4）

- **记背** WSS = 均值常数 + $R_X(t,t+\tau)=R_X(\tau)$；$R(0)=E[X_t^2]\ge|R(\tau)|$；$R(-\tau)=R(\tau)$（实）；$X_t=A\cos(\omega t+\Theta)$、$\Theta\sim U[0,2\pi)$⇒WSS 且 $R(\tau)=\frac{A^2}{2}\cos\omega\tau$；均值遍历⇔$\frac{1}{T}\int_0^T(1-\frac{\tau}{T})C(\tau)d\tau\to 0$（离散 $\frac{1}{n}\sum_k(1-\frac{k}{n})C(k)\to 0$）。

考点 11 | 泊松过程（课件 P4）

- **记背** $P\{N(t)=k\}=e^{-\lambda t}(\lambda t)^k/k!$；独立平稳增量；$E[N(t)]=\mathrm{Var}(N(t))=\lambda t$；间隔 T_i i.i.d. $\mathrm{Exp}(\lambda)$（无记忆）；独立叠加 = 泊松（速率相加）；独立稀释（概率 p）= 泊松(λp) 且独立；给定 $N(t)=n$ 到达时刻 = n 个 i.i.d. $U[0,t]$ 的顺序统计量。

考点 12 | 马尔可夫链（课件 P4）

- **记背** 马氏性 $P\{X_{n+1}|X_0..X_n\}=P\{X_{n+1}|X_n\}$；转移矩阵行和 1；C-K $P^{(n+m)}=P^nP^m$；分布演化 $\mu_{n+1}=\mu_nP$；平稳分布 $\pi P=\pi$、$\sum\pi_i=1$；两态 $\begin{pmatrix}1-a&a\\b&1-b\end{pmatrix}\Rightarrow\pi=(\frac{b}{a+b},\frac{a}{a+b})$。

通用记背 | 分布 · 积分 · 不等式 · 矩阵 · 矩

- **分布（均值/方差/特征函数）**：$\mathrm{Be}(p)$：p，$p(1-p)$，$1-p+pe^{ju}$；$\mathrm{Bin}(n,p)$：np，$np(1-p)$，$(1-p+pe^{ju})^n$；$\mathrm{Pois}(\lambda)$：λ，λ，$e^{\lambda(e^{ju}-1)}$；$U[a,b]$：$\frac{a+b}{2}$，$\frac{(b-a)^2}{12}$，$\frac{e^{jub}-e^{jua}}{ju(b-a)}$；$\mathrm{Exp}(\lambda)$：$\frac{1}{\lambda}$，$\frac{1}{\lambda^2}$，$\frac{\lambda}{\lambda-ju}$；$N(\mu,\sigma^2)$：$\mu$，$\sigma^2$，$e^{ju\mu-\sigma^2u^2/2}$。
- **积分/极限**：$\int_{-\infty}^{\infty}e^{-ax^2+bx}dx=\sqrt{\pi/a}\,e^{b^2/(4a)}$；$\int_0^\infty x^ne^{-ax}dx=n!/a^{n+1}$；高斯×高斯⇒精度相加；$(1+z_n/n)^n\to e^z$；Taylor：$e^x=1+x+\frac{x^2}{2}$，$\cos x=1-\frac{x^2}{2}$，$\ln(1+x)=x-\frac{x^2}{2}$。
- **三角正交**：$\Theta\sim U[0,2\pi)$：$E[\cos n\Theta]=0$、$E[\cos^2 n\Theta]=\frac{1}{2}$、$E[\cos n\Theta\cos m\Theta]=0$（$n\ne m$）；$n\Theta \bmod 2\pi\sim U[0,2\pi)$；积化和差。
- **不等式**：Markov、Chebyshev、Chernoff、Cauchy–Schwarz $|E[XY]|\le\sqrt{E[X^2]E[Y^2]}$、Jensen（凸）、$(a+b)^2\le 2a^2+2b^2$。
- **矩阵/矩**：2×2 逆 $\frac{1}{ad-bc}\begin{pmatrix}d&-b\\-c&a\end{pmatrix}$；协方差对称半正定（特征值/主子式≥0）；Cholesky 对角元 = 新息方差；$M_Z^{(k)}(0)=E[Z^k]$；$\mathrm{Cov}(AX+b,CY+d)=A\mathrm{Cov}(X,Y)C^T$；$\mathrm{Var}=E[X^2]-(E[X])^2$。

反例库（收敛关系"反之不成立"）

- **a.s.⇏m.s.**：$X_n=n\cdot\mathbf{1}_{(0,1/n]}$：a.s.→0 但 $E[X_n^2]=n\to\infty$。
- **m.s./p⇏a.s.**：滑动 blips $\mathbf{1}_{I_n}$，$|I_n|\to 0$ 反复扫过：m.s.、p→0 但不 a.s.（每点无限次命中）。
- **d⇏p**：$X\sim N(0,1)$、$X_n=-X$：同分布⇒d 收敛，但 $X_n-X=-2X\not\to 0$。
- **p⇏m.s.（无界）**：$X_n=n$ w.p. $1/n$：p→0 但 $E[X_n^2]=n\to\infty$。
- **不连续映射**：$X_n=(-1)^n/n\to 0$，阶跃 $h\Rightarrow h(X_n)=(-1)^n$ 不收敛。
- **有界救场**：$|X_n|\le C$ 且 p 收敛⇒m.s. 收敛。

证明手法速查

- **证 a.s.**：Borel–Cantelli（$\sum P<\infty$）/ 逐点分析 / 单调有界 / 连续映射。
- **证 m.s.**：直接二阶矩 / Cauchy 准则 / 控制收敛交换极限。
- **证 p**：定义 + Markov/Chebyshev / 借 m.s. 或 a.s.。
- **证 d**：Lévy 特征函数 / CDF 连续点 / 同分布 / CLT·Poisson 模板。
- **证不收敛**：增量不趋于 0 / Cauchy 失败 / 特征函数振荡或极限在 0 不连续 / 有界 +p⇒m.s. 矛盾 / $(-1)^n$ 振荡。